

# Resource-Constrained Neural Monte Carlo Tree Search for Quantum Boolean Oracle Synthesis

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## Abstract

Boolean oracle synthesis embeds classical predicates into quantum algorithms, but direct algebraic-normal-form constructions can be costly in non-Clifford gates, CNOTs, logical depth, and temporary workspace. We formulate bit-flip Boolean oracle synthesis as a resource-constrained search problem over ANF/FPRM term sets and introduce Resource-NMCTS, a logical-layer workflow that combines neural action priors, Monte Carlo tree search, Boolean-ring actions, frontier controllers, Pareto archives, and baseline-preserving guards. Unlike SSHR, the method does not search parallelotope candidates; SSHR is used only as a CNOT-oriented small-function baseline. On 177 traditional functions with  $n \leq 6$ , Pareto-Resource-NMCTS reduces mean T-count by 72.25% and mean weighted score by 67.80% relative to direct ANF synthesis, and its score win/loss/tie counts against ESOP beam, weighted ESOP-MILP, and SSHR-H are 174/0/3, 167/3/7, and 173/4/0. External probes show corresponding score wins against ROS-style LUT (309/0/0), mockturtle (166/11/0 and 64/0/0), CirKit (177/0/0 and 64/0/0), and legacy RevKit CLI exact-oracle synthesis (173/0/4). These comparisons also expose the expected tradeoffs: CirKit is often shallower, and RevKit uses fewer auxiliary lines on most small functions. In the phase branch, a rank-trained shortlist scores 512 of 8192 affine forms per held-out  $n = 6$  function and matches the wide-search mean within 0.01% under the  $T/R_z = 30$  proxy. For larger term-set instances, a sparse depth-2/4 frontier matches the measured depth-2/3/4 frontier while reducing evaluation time by 24.94–28.88%, and a learned depth-4 gate preserves sparse-frontier score on a 144-pair multi-seed  $n = 24, 28, 32, 40$  audit while saving 13.43%. Correctness is supported by 15,294 symbolic, phase-search, and policy-pruning rows, plus 400/400 complete truth table-bridge rows for  $n = 21$ –25. The contribution is a verified logical-layer search method with strong T-count and weighted-score advantages, not a hardware-mapped, depth-only, or universal-dominance claim.

## 1 Introduction

Given a Boolean function  $f : \{0, 1\}^n \rightarrow \{0, 1\}$ , a bit-flip oracle implements

$$|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle. \quad (1)$$

Such oracles appear in Grover search, constraint solving, and quantum cryptanalysis. Their cost is not determined only by the number of Boolean operations. In a fault-tolerant or resource-limited setting, the chosen Boolean representation, compute/uncompute order, temporary workspace, and phase realization all affect T-count, CNOT count, logical depth, gate count, and peak ancilla.

Existing synthesis routes provide strong local solutions. ESOP, XAG, LUT and BDD representations reduce different Boolean resources; ROS makes LUT-based oracle synthesis resource-aware; RevKit, mockturtle, CirKit, and ABC provide mature logic and reversible-synthesis toolchains; and SSHR uses parallelotope structures in the Boolean hypercube to target small-scale CNOT-oriented synthesis [6, 5, 4, 15, 1, 13, 12, 17, 11, 10, 9]. These methods motivate

the baselines used in this paper, but they also expose a gap: a fixed representation or a single resource objective may miss useful tradeoffs when T-count, CNOTs, depth, gates, and ancilla are optimized together.

We address this gap by treating Boolean oracle synthesis as an explicit search problem. The state is an ANF/FPRM term set; actions are algebraic rewrites that can be expanded back to GF(2) semantics; neural policies and MCTS allocate the search budget; and guard rules ensure that learned candidates do not replace a stronger deterministic baseline. The resulting circuit is checked at the plan level, at the emitted X/CNOT/MCT symbolic level, and, for bridge instances, by full truth table oracle simulation.

The paper makes four contributions. First, it defines Resource-NMCTS as a resource-constrained ANF/FPRM search pipeline rather than as an SSHR extension. Second, it combines Boolean-ring linear-factor screening, neural action priors, MCTS, depth-frontier policies, Pareto archives, and baseline-preserving guards into one logical-layer synthesis workflow. Third, it evaluates the method against ESOP, SSHR, ABC/BDD, ROS-style LUT, mockturtle, CirKit, RevKit CLI, and RevKit phase/Rz baselines on matched functions. Fourth, it separates algorithmic gains from engineering guards through ablations and high-dimensional verification bridges.

**Scope and assumptions.** The study is intentionally limited to logical-layer Boolean-oracle synthesis. All reported resources are computed before hardware mapping, routing, native-gate scheduling, noise modeling, and magic-state-factory accounting. The weighted score in Eq. (3) is used only to rank verified candidates; T-count, CNOT count, depth, gate count, and peak ancilla are also reported separately, and cross-toolchain comparisons are interpreted through the baseline comparability audit rather than as a universal leaderboard.

Table 1: Contribution-to-evidence map. Each contribution is tied to concrete implementation mechanisms, paper evidence, and the boundary that prevents the claim from being overstated.

| Contribution                                            | Implementation                                                                                                                       | Paper evidence            | Boundary                                                                                                                                    |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Resource-constrained Boolean-oracle search formulation  | ANF/FPRM term-set state, Boolean-ring and affine actions, guarded candidate selection, symbolic circuit emission.                    | Fig. 1; Table 2           | Logical-layer oracle synthesis only; no routing, native-gate mapping, or noise model.                                                       |
| Neural/MCTS resource-search workflow                    | Neural action priors, MCTS/beam/portfolio search, Pareto archive, frontier policies, staged and sparse learned gates.                | Tables 21, 22, 28; Fig. 5 | The largest gains come from searchable algebraic actions plus guarded selection; the paper does not claim deep RL alone explains all gains. |
| Broad baseline and toolchain comparison                 | Direct ANF, ESOP beam/MILP, SSHR, ABC/BDD, ROS-style LUT, mockturtle, CirKit, RevKit CLI, and phase/Rz probes.                       | Tables 4, 5, 17; Fig. 3   | Strong T-count and weighted-score evidence, not universal CNOT/depth/ancilla dominance or full ROS reproduction.                            |
| High-dimensional and phase-search verification envelope | $n = 20$ –40 symbolic term-set checks, $n = 21$ –25 complete truth-table bridges, Affine-FPRM phase search, learned phase shortlist. | Tables 34, 35, 33; Fig. 7 | Large rows are logically and symbolically verified; complete truth-table enumeration remains limited to bridge slices.                      |

**a** Resource-constrained neural MCTS oracle synthesis remains verifiable at every layer.

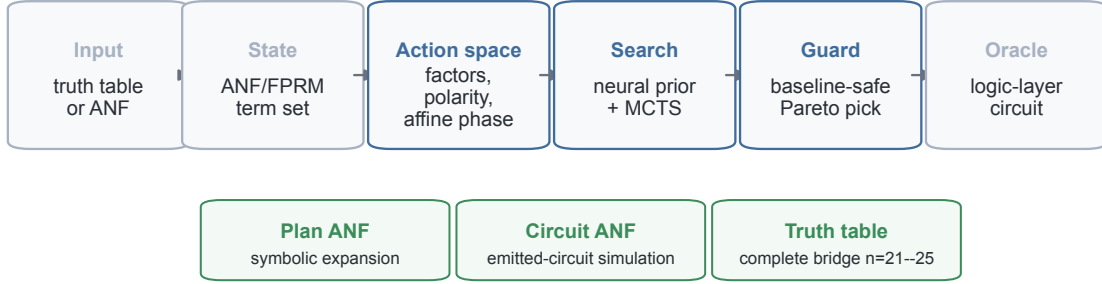


Figure 1: **Verified search pipeline.** Resource-NMCTS operates on complete truth table inputs or ANF term sets, searches algebraic actions under neural and tree-search control, and verifies candidate circuits at plan, emitted symbolic-circuit, and complete-truth table levels.

## 2 Related Work

### 2.1 Boolean representations for oracle synthesis

Low-T oracle synthesis is closely tied to the Boolean intermediate representation. BDD-based reversible synthesis uses symbolic decision diagrams to handle large functions [15]. ROS frames oracle synthesis as resource-constrained LUT mapping with downstream reversible implementation costs [6]. XAG-based compilation separates XOR structure from AND nodes, making multiplicative complexity a proxy for non-Clifford cost [5, 4]. Back-end-aware oracle synthesis further connects logic synthesis to downstream resource metrics, but that line includes mapping concerns that are outside the scope of this logical-layer study [16].

### 2.2 Reversible and logic-synthesis toolchains

ABC supplies mature AIG, XAG/GIA, LUT, and ESOP optimization flows for classical logic [1]. mockturtle and CirKit provide reusable logic-network optimization libraries and shells in the EPFL logic-synthesis ecosystem [13, 10, 9]. RevKit provides reversible-logic and quantum-compilation functionality, including the Python `oracle_synth` API and legacy command-line reversible synthesis flows [12, 11]. In this work these toolchains are used as external baselines or probes. They are not called inside Resource-NMCTS, which keeps the proposed search policy separated from the comparison backends.

### 2.3 Learning-guided synthesis

Learning-guided synthesis has been explored for both classical and quantum circuits. ShortCircuit uses AlphaZero-style search for classical Boolean circuit design [14]. Gumbel AlphaZero has been used for Clifford+T unitary synthesis [7]; reinforcement learning has been used for non-Clifford-count optimization and ZX-calculus rewriting [2, 8]; MCTS has been used for quantum circuit transformation under connectivity constraints [18]; and MonteQ uses MCTS for Hamiltonian-simulation circuit synthesis [3]. The present work is different in scope: it is a logical-layer Boolean-oracle synthesizer, not a hardware-routing method, not a Hamiltonian-simulation scheduler, not a gate-by-gate unitary generator, and not an SSHR parallelootope search. Its learned component ranks verifiable Boolean-algebraic actions, while ESOP, BDD, XAG/LUT, ROS-style, SSHR, RevKit, mockturtle, CirKit, and ABC flows define the comparison set.

Table 2 summarizes this positioning. It is used to keep the manuscript’s contribution narrow and testable: existing tools and representations define strong baselines or motivation, while Resource-NMCTS is claimed only as a logical-layer resource-aware search method over verifiable Boolean-algebraic actions.

Table 2: Related-work positioning matrix. The table states what each literature family optimizes, which gap motivates the proposed search, and how the manuscript uses that family in experiments or claim boundaries.

| Line of work                      | Representative work                                                                   | Main lever                                                                 | Gap for this paper                                                                                                                                                       | Manuscript use                                                                                        |
|-----------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| BDD-based reversible synthesis    | BDD reversible synthesis [15]                                                         | Symbolic decision-diagram representation for large Boolean functions.      | Representation scalability does not by itself optimize T, CNOT, depth, gates, and ancilla jointly.                                                                       | Used as a representation-level baseline family, not as the proposed search space.                     |
| LUT/ROS-style oracle synthesis    | ROS and back-end-aware oracle synthesis [6, 16]                                       | Resource-aware LUT mapping and downstream implementation-cost estimates.   | Full ROS garbage-management and hardware/back-end mapping are outside the logical-layer claim.                                                                           | Reproduced through a verified ROS-style LUT proxy and line-aware reselection stress tests.            |
| XAG and multiplicative complexity | XAG compilation and multiplicative-complexity oracle cost [4, 5]                      | Separates XOR-linear structure from AND nodes as a non-Clifford proxy.     | A fixed network optimization flow may miss algebraic FPRM/ANF search actions and Pareto tradeoffs.                                                                       | Used through ABC, mockturtle, and CirKit probes plus exact small XAG lower-bound checks.              |
| Logic and reversible toolchains   | ABC, EPFL libraries, RevKit, mockturtle, and CirKit [1, 13, 12, 11, 10, 9]            | Mature optimization passes and reversible-synthesis flows.                 | Tool output defines strong baselines but does not expose the neural/MCTS algebraic action policy.                                                                        | Used as external probes and exact-oracle reversible-synthesis comparisons.                            |
| SSHR geometric synthesis          | Parallelotope-based CNOT-oriented synthesis [17]                                      | Small-scale Boolean hypercube geometry targeting CNOT-oriented covers.     | The objective is CNOT-oriented and small-function focused, not a general resource-weighted AI search.                                                                    | Used as a CNOT-oriented small-function baseline; not used by the proposed method.                     |
| Learning-guided circuit synthesis | AlphaZero, reinforcement learning, MCTS, and ZX-guided synthesis [14, 7, 2, 8, 18, 3] | Learns or searches over circuit-design and circuit-transformation actions. | Most targets are classical circuits, unitary synthesis, routing/transformation, ZX rewriting, or Hamiltonian simulation rather than Boolean-oracle ANF/FPRM term search. | Motivates neural/MCTS control while keeping actions verifiable at the Boolean-algebraic oracle layer. |

### 3 Problem and Resource Model

We represent a Boolean function by its square-free ANF

$$f(x) = \bigoplus_{m \in \mathcal{M}} \prod_{i \in m} x_i, \quad (2)$$

where each monomial  $m$  is stored as a bit mask. A direct construction applies one controlled action per monomial. The synthesis problem is to replace this direct construction with reusable compute/action/uncompute subplans that implement Eq. (1) while lowering a multi-resource objective.

All resource numbers in this paper are logical-layer estimates over X, CNOT and multi-controlled Toffoli gates. Candidate selection uses

$$\text{score} = T + 0.04 \text{ CNOT} + 0.015 \text{ depth} + 0.01 \text{ gates} + 2 \text{ ancilla}. \quad (3)$$

The score is a ranking objective, not a physical runtime or error model. We therefore report T-count, CNOT count, depth, gate count, and peak ancilla separately whenever a tradeoff is relevant.

## 4 Method

### 4.1 Search state and actions

The Resource-NMCTS state is the current set of unsynthesized ANF terms. An action chooses a reusable algebraic structure, synthesizes that structure into a temporary line, uses it to control a quotient subproblem, and then uncomputes it. Candidate actions include common monomial factors, FPRM polarity rewrites, CNOT-computed linear parity factors, Boolean-ring linear factors, and bounded affine changes of variables. Each action is scored by its immediate resource estimate and by the size and structure of the induced subproblems.

Table 3 summarizes the complete execution path. The central invariant is that learning and tree search only control which verifiable algebraic candidates are evaluated or selected; semantic correctness is supplied by the ANF, emitted-circuit, truth-table, or phase verifier attached to the corresponding stage.

Table 3: End-to-end synthesis and verification workflow. Each stage records the mechanism used by Resource-NMCTS, the invariant that prevents the learned search from becoming a semantic oracle, and the artifact produced for downstream analysis.

| Stage                      | Mechanism                                                                                                                                                     | Invariant                                                                                                          | Artifact                                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Input normalization        | Convert a truth table, exported benchmark, or supplied term set into a square-free ANF monomial mask set $\mathcal{M}$ .                                      | The term set must evaluate to the target Boolean function before any search step is trusted.                       | ANF terms, preset name, and source manifest.                                                                 |
| Candidate generation       | Generate direct ANF, FPRM polarity, common-factor, Boolean-ring linear-factor, affine, and depth-frontier candidates.                                         | Every action must have an explicit GF(2) expansion back to the original variables.                                 | Candidate logical plans with resource estimates.                                                             |
| Search control             | Use neural action priors, MCTS/beam expansion, staged frontiers, and sparse learned gates to allocate the search budget.                                      | Learned decisions rank or skip evaluations; they do not define semantic correctness.                               | Scored candidate pool and search-time diagnostics.                                                           |
| Guarded selection          | Compare learned and expensive candidates with deterministic baselines, then apply Pareto filtering before active-score selection.                             | A selected plan must be a valid generated candidate and remains bounded by the stated guard and score model.       | Selected logical plan and non-dominated archive entry.                                                       |
| Circuit emission           | Expand the selected plan into X/CNOT/MCT compute-action-uncompute operations and compute logical resource counts.                                             | The reported score is a ranking objective; T, CNOT, depth, gates, and ancilla are retained separately.             | Logical circuit proxy and resource row.                                                                      |
| Verification and reporting | Check the plan algebraically, check emitted-circuit ANF semantics, add full truth-table or phase checks where feasible, and write raw CSV plus manifest rows. | Paper comparisons use matched rows that pass the relevant semantic checks and exclude errors, skips, and timeouts. | <code>raw*.csv</code> ,<br><code>summary*.csv</code> ,<br><code>manifest*.json</code> , and<br>paper tables. |

### 4.2 Boolean-ring linear factors

A key extension is allowing the linear factor and the quotient to share variables under Boolean-ring semantics. For example,

$$(x_i \oplus x_j)x_im = x_im \oplus x_ix_jm, \quad (4)$$

because  $x_i^2 = x_i$  over the Boolean ring. Circuit-wise, the parity  $x_i \oplus x_j$  is computed by CNOTs into a temporary line, used as a control for the quotient plan, and uncomputed. This expands the algebraic action space without changing the GF(2) verifier.

### 4.3 Neural prior, MCTS, frontier policy, and guard

The neural action prior ranks candidate actions using term-count, degree, coverage, variable-support, and immediate-resource features. MCTS then explores combinations of actions with deterministic rollout estimates. A separate frontier policy selects among depth-2, depth-3, and depth-4 Boolean screening frontiers; a cost-aware version trades a small score increase for lower planning time and reduced ancillary lifetime area. We also use two conditional frontier controllers. The staged controller evaluates depth-2 and depth-3 screens first and only runs depth-4 when the measured depth-3 improvement is large enough. The sparse depth-4 gate instead uses the depth-2 state to predict whether the deterministic sparse depth-2/4 frontier needs its depth-4 evaluation; its threshold is selected on validation data to avoid false skips of improving depth-4 cases. Finally, a guard compares learned or expensive candidates to deterministic baselines and returns the best valid candidate under Eq. (3). This makes learning a search-control component, not the source of semantic correctness.

### 4.4 Pareto archive

The Pareto variant evaluates candidates under several internal objectives, including active-score, T-sparse, CNOT/depth-heavy, gate-sensitive, and ancilla-tight rankings. It removes candidates dominated simultaneously in T-count, CNOT count, depth, gate count, and peak ancilla, then selects from the remaining archive using the active score. The archive is used only after each candidate has passed the same plan and emitted-circuit checks as the base method.

### 4.5 Phase/Rz branch

RevKit `oracle_synth` returns phase/Rz netlists rather than direct bit-flip Clifford+T circuits. To avoid treating that boundary only as a cost sensitivity, we also implement a phase-parity emitter: each ANF monomial is expanded into merged parity-phase gadgets and verified up to global phase. Fixed-polarity FPRM search and bounded affine preconditioning are then added as phase-search actions. This branch is reported as a logical phase/Rz proxy; it does not output approximate rotation sequences or mapped hardware circuits.

## 5 Experimental Design

Experiments answer four questions. First, does Resource-NMCTS reduce T-count and weighted score on small functions where full truth table checking is practical? Second, do the conclusions survive comparisons with ESOP, SSHR, ABC/BDD, ROS-style LUT, mockturtle, CirKit, RevKit CLI, and RevKit phase/Rz baselines? Third, which components—affine search, MCTS, neural priors, guards, frontier policies, and Pareto archives—account for the gains? Fourth, how far can correctness evidence be pushed beyond small truth tables?

The traditional slice contains 177 functions with  $n \leq 6$ . High-dimensional logic-network probes include exported  $n = 14$ ,  $n = 15$ ,  $n = 16$ , and  $n = 18$  random-ANF sets for ABC/BDD, and  $n = 14$  sets for mockturtle and CirKit. Scale and bridge experiments cover item-level  $n = 20$ –40 term sets and complete truth table bridge functions at  $n = 21$ –25. Paired comparisons include only matching functions or items with successful semantic checks and no timeout. Because these baselines operate at different abstraction levels, Table 4 first separates their role in the argument from the claims they cannot support. Table 5 then summarizes the quantitative comparison evidence for each baseline family.

Table 4: Baseline claim matrix. The table explains why each comparison family is included, what conclusion it supports, and which stronger conclusion is outside the scope of the logical-layer experiments.

| Role                                  | Compared methods                                                                                          | Why it matters                                                                                                                | Supported claim                                                                                                              | Excluded claim                                                                                |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Primary resource-efficiency baselines | Direct ANF, AND-direct ANF, ESOP beam/MILP, BDD/ABC-ESOP, SSHR-H/SSHR-I                                   | These methods solve the same bit-flip Boolean-oracle task under the paper’s logical resource model.                           | Resource-NMCTS improves T-count and weighted score on matched small functions while SSHR remains a strong CNOT counterpoint. | Does not prove universal CNOT, depth, ancilla, or hardware-level optimality.                  |
| External logic-network probes         | ABC AIG/XAG/LUT, ROS-style LUT proxy, mockturtle KLUT-to-XAG, CirKit AIG/MC                               | They test whether mature Boolean-network optimization already removes the apparent advantage over direct algebraic baselines. | The score advantage persists against stronger exported-network and official-header tool probes.                              | Does not reproduce full ROS SAT garbage management or reversible/hardware mapping.            |
| Exact reversible-synthesis probe      | Legacy RevKit CLI TBS/DBS/RMS exact-oracle portfolio                                                      | It embeds each function as $(x, y) \mapsto (x, y \oplus f(x))$ and checks a genuine reversible-synthesis toolchain.           | Resource-NMCTS has lower T-count and weighted score against this portfolio but pays more auxiliary lines.                    | Does not imply lower line count, routed depth, or mapped Clifford+T cost.                     |
| Phase/ $R_z$ branch                   | RevKit oracle_synth, phase-parity ANF, fixed-polarity FPRM, affine FPRM, learned phase shortlist          | It tests whether the same search framing extends to phase-oracle and affine-preconditioned $R_z$ cost proxies.                | Affine and learned candidate pruning improve verified logical phase/ $R_z$ proxy objectives.                                 | Does not output approximate rotation sequences or a final Clifford+T decomposition.           |
| Internal search ablations             | No-MCTS portfolio, learned prior off, guard off, Pareto archive off, depth-frontier variants, sparse gate | They separate representation effects from the contribution of search, neural ranking, guards, and budget control.             | AI-guided search and guards add measurable gains or time savings beyond deterministic construction.                          | Does not support a claim that deep reinforcement learning alone causes the whole improvement. |
| Scaling and correctness bridges       | $n = 20\text{--}40$ term-set symbolic runs; $n = 21\text{--}25$ complete truth-table bridge rows          | They push semantic checks beyond the small truth-table slice while keeping full verification where still feasible.            | The emitted logical-layer circuits remain symbolically and bridge-truth-table verified at larger dimensions.                 | Does not make exhaustive high-dimensional truth-table benchmarking feasible or complete.      |
| Trade-off audits                      | Raw multi-resource dominance, line-aware LUT reselection, schedule proxy, sparse-threshold sweep          | They prevent a weighted-score result from hiding CNOT, depth, line-count, or planning-time tradeoffs.                         | The method occupies a strong T/score point with explicit depth, CNOT, ancilla, and runtime boundaries.                       | Does not justify reporting a single-metric victory as complete dominance.                     |

Table 5: Reviewer-facing evidence matrix. The table consolidates the baseline family, coverage, verification count, headline paired result, and scope boundary used by the manuscript claims.

| Evidence block                   | Scope                     | Verified rows                           | Main score result                                                                             | Boundary                                                                                                                          |
|----------------------------------|---------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Internal logical baselines       | $n = 3-6$                 | 1770/1770                               | Pareto vs direct ANF 172/1/4, -67.80%; vs ESOP-MILP 167/3/7, -29.84%                          | Same verifier/cost model; includes direct, AND-direct, ESOP beam/MILP, SSHR-H, MCTS, affine, and Pareto variants.                 |
| Exported SSHR/ABC/BDD extension  | $n = 3-6$                 | 1416/1416                               | Resource-NMCTS vs SSHR-I CNOT 172/5/0, -29.48%; vs ABC-XAG 177/0/0, -64.15%                   | SSHR-I rows use an 8 s Gurobi budget; ABC/BDD rows are logical estimates over exported truth tables.                              |
| ROS-style LUT proxy              | $n = 3-6, 14, 15, 16, 18$ | 309/309 best rows; 927/927 K-sweep rows | Pareto vs best-K proxy 309/0/0, -84.27%                                                       | Verified LUT proxy only; no official ROS SAT garbage management, reversible emission, or hardware mapping.                        |
| mockturtle KLUT-to-XAG probe     | $n = 3-6, 14$             | 241/241                                 | $n \leq 6$ 166/11/0, -31.50%; $n = 14$ 64/0/0, -91.49%                                        | Official-header XAG resynthesis probe; still a logical proxy, not full ROS or reversible garbage management.                      |
| CirKit AIG/MC probe              | $n = 3-6, 14$             | 241/241                                 | $n \leq 6$ 177/0/0, -62.34%; $n = 14$ 64/0/0, -94.46%                                         | AIG multiplicative-complexity probe; strongest current depth-oriented external counterpoint.                                      |
| Legacy RevKit CLI exact oracle   | $n = 3-6$                 | 708/708                                 | Pareto vs best-score portfolio 173/0/4, -67.28%                                               | Exact reversible oracle permutation; CNOT/depth are derived from Toffoli-control distributions and ancilla is a visible tradeoff. |
| RevKit phase/Rz branch           | $n = 3-6$                 | 531/531                                 | Affine phase vs RevKit oracle_synth 177/0/0, -65.50%                                          | Logical phase/Rz proxy verified up to global phase; not an approximate-rotation or final Clifford+T sequence.                     |
| Learned phase pruning            | $n = 3-6$                 | 7611/7611                               | Held-out $n=6$ rank-diverse top512 vs wide128 0/7/31, +0.00%                                  | Policy-pruned affine candidate ranking; evidence is held-out exact scoring, not a standalone compiler backend.                    |
| High-dimensional frontier search | $n = 20, 24, 28, 32, 40$  | 1608/1608                               | Stage-gated vs all-depth scale 0/4/92, +0.04% score; -25.43% time                             | Term-set symbolic verification with emitted-circuit ANF checks; depth frontier is a planning guard, not a hardware scheduler.     |
| Complete truth-table bridges     | $n = 21-25$               | 400/400                                 | Bridge rows verify plan, emitted symbolic circuit, and complete truth table for $n = 21-25$ . | Bridge set is intentionally narrow because complete truth tables grow exponentially.                                              |

Table 6 makes the comparison protocol explicit as reviewer-facing questions. It is included to prevent the baseline set from being read as a single leaderboard: each layer has a usable conclusion and a stronger conclusion that the evidence does not support.

Table 6: Comparison protocol audit. Each comparison layer is tied to the reviewer question it answers, the conclusion supported by the current evidence, and the stronger claim excluded by the logical-layer experiment design.

| Comparison layer                              | Reviewer question                                                                          | Usable conclusion                                                                                                       | Excluded conclusion                                                                                      |
|-----------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Primary same-task Boolean-oracle baselines    | Is the method only compared with weak or self-written constructions?                       | Matched bit-flip oracle baselines support lower T-count and weighted score under the paper’s logical model.             | They do not prove universal CNOT, depth, ancilla, line-count, or hardware-mapped optimality.             |
| External logic-network and LUT/XAG/AIG probes | Does the advantage survive mature external logic-synthesis toolchains?                     | External probes test whether the score advantage persists beyond the project’s internal baselines.                      | They are not a full ROS SAT garbage-management flow, reversible emission, or hardware-mapped comparison. |
| Exact reversible-synthesis counterpoint       | Is there a reversible-synthesis comparison, not only logic-network proxies?                | RevKit CLI supports a genuine exact-oracle reversible-synthesis probe with lower T/score for Resource-NMCTS.            | It does not imply lower auxiliary-line count, routed depth, or mapped Clifford+T cost.                   |
| Phase/ $R_z$ proxy branch                     | How is the RevKit phase/ $R_z$ gate-set mismatch handled?                                  | The phase branch supports a logical $R_z$ /phase proxy and learned affine shortlist result.                             | It is not a final approximate-rotation sequence or Clifford+T decomposition.                             |
| Internal search-control ablations             | Is the AI/MCTS contribution separated from the algebraic search space?                     | Ablations support bounded search-control, pruning, gating, and planning-time gains.                                     | They do not show that deep reinforcement learning alone causes the main resource drop.                   |
| Scaling and correctness bridges               | Does the work go beyond small truth-table functions while preserving correctness evidence? | Large rows support logical symbolic correctness and bridge-truth-table verification within the stated envelope.         | They do not prove exhaustive high-dimensional truth-table optimality or benchmarking completeness.       |
| Trade-off and non-dominance audits            | Does the weighted score hide unfavorable CNOT, depth, ancilla, or runtime behavior?        | The paper can claim a strong logical T/weighted-score point while explicitly reporting unfavorable resource dimensions. | It cannot report a single-metric score win as complete Pareto or hardware dominance.                     |

Table 7 makes the fairness boundary explicit. It separates comparisons that share the same bit-flip oracle target from logic-network proxies, exact reversible-oracle probes, phase/ $R_z$  proxies, and large symbolic bridge checks. The table is intentionally conservative: a comparison is used only for the conclusion supported by its task alignment, verification route, and resource abstraction.

Table 7: Baseline comparability audit. The table records why each comparison family is fair enough for the paper’s bounded claim, which controls are used, and what residual abstraction mismatch remains.

| Baseline family                              | Task alignment                                                                                                              | Fairness control                                                                                                                               | Residual risk                                                                                                        | Usable claim                                                                                                                          |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Direct algebraic and ESOP baselines          | Same bit-flip Boolean oracle target and the same logical resource model.                                                    | Matched $n \leq 6$ functions, complete truth-table checks, identical score coefficients, and paired per-function comparisons.                  | These baselines are representation-level constructions, not mature external compilers.                               | Primary evidence for lower T-count and weighted score under the paper’s own oracle model.                                             |
| SSHR-H and SSHR-I                            | Same small Boolean-function oracle setting, but with a CNOT-oriented parallelotope search objective.                        | Matched functions and resource extraction; SSHR-I timeout rows are separated from completed rows.                                              | SSHR is optimized for CNOT structure and small $n$ , so it is a counterpoint rather than the method’s design parent. | Resource-NMCTS improves score and T-count while SSHR remains the strongest CNOT reference.                                            |
| ABC/BDD and exported logic networks          | Same exported Boolean functions, evaluated through logical AIG/XAG/LUT/BDD resource proxies.                                | Rows with errors, skips, or failed truth-table checks are excluded; paired comparisons use matched names.                                      | Network-level counts are proxies and do not include reversible garbage management or quantum routing.                | Mature classical logic optimization does not by itself remove the T/score advantage.                                                  |
| ROS-style LUT, mockturtle, and CirKit probes | Same benchmark functions are passed through LUT/XAG/AIG toolchain probes and verified after export/readback when available. | The LUT proxy includes best-K and line-aware reselection; mockturtle and CirKit use official toolchain paths and row-level correctness checks. | These are not full ROS SAT garbage-management or hardware-mapped reversible flows.                                   | The advantage persists beyond self-written baselines, with CirKit exposing a real depth tradeoff.                                     |
| Legacy RevKit CLI exact-oracle portfolio     | Each function is embedded as the exact reversible permutation $(x, y) \mapsto (x, y \oplus f(x))$ .                         | TBS, DBS, and RMS are all run and reduced to a per-function best-score portfolio.                                                              | CNOT and depth are derived from Toffoli-control distributions; RevKit uses fewer auxiliary lines on most rows.       | A genuine reversible-synthesis probe still favors Resource-NMCTS on T-count and score, but not on line count.                         |
| Phase/ $R_z$ baselines and learned shortlist | Same Boolean functions are evaluated as phase-oracle or affine-FPRM phase-search instances.                                 | Rows are verified up to global phase; learned shortlists are checked against same-budget random controls and wide exact search.                | The branch is a logical $R_z$ /phase proxy and does not output rotation-synthesis sequences.                         | The search framing extends to verified phase/ $R_z$ proxies, but it is not a finished Clifford+T compiler.                            |
| High-dimensional symbolic and bridge checks  | Large term-set instances and $n = 21$ –25 bridge functions test the same emitted logical oracle semantics.                  | Symbolic ANF/circuit checks cover all reported high-dimensional rows; bridge slices add complete truth-table checks.                           | Complete truth-table enumeration is limited to bridge slices and generated functions.                                | The large-scale evidence supports semantic correctness and scaling of the logical search, not exhaustive high-dimensional optimality. |

Table 8 records the strongest opposing evidence that bounds the comparison claims. This audit is included because a meaningful comparison should not hide metrics on which other methods remain strong. SSHR is therefore treated as a CNOT counterpoint, CirKit as a depth counterpoint, RevKit CLI as an auxiliary-line counterpoint, and the learned-prior rows as evidence for incremental search control rather than for a deep-RL-only explanation.

Table 8: Counterpoint and claim-boundary audit. The table reports the strongest unfavorable metric evidence that constrains the manuscript’s claims, the favorable evidence that keeps each comparison meaningful, and the resulting claim boundary.

| Counterpoint                        | Opposing evidence                                                        | Favorable evidence                                                      | Claim boundary                                                                                                              |
|-------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| SSHR family CNOT optimum            | vs SSHR-I CNOT: CNOT 0/168/9, +62.06%; vs SSHR-H: CNOT 43/128/6, +18.99% | score vs SSHR-I CNOT 173/4/0, -31.89%; score vs SSHR-H 173/4/0, -41.06% | Use SSHR as a small-function CNOT counterpoint; claim T-count and weighted-score advantage, not CNOT dominance.             |
| CirKit AIG/MC depth                 | depth 16/156/5, +93.16%                                                  | score 177/0/0, -62.34%                                                  | Use CirKit as a depth-oriented external probe; do not claim depth dominance over logic-network synthesis.                   |
| RevKit exact-oracle auxiliary lines | peak ancilla 0/169/8, +153.11%                                           | score 173/0/4, -67.28%                                                  | Use RevKit CLI as an exact reversible-oracle probe; do not claim line-count or mapped Clifford+T dominance.                 |
| Learned prior is incremental        | time 37/140/0, +18.77% versus no-prior rerun                             | score 29/0/148, -0.78%                                                  | Frame AI as search control, ranking, pruning, and gating; do not claim deep RL alone explains the main resource drop.       |
| Large- $n$ verification is bounded  | complete truth-table enumeration is limited to bridge slices             | symbolic scale rows 1608/1608; complete truth-table bridge rows 400/400 | Claim logical correctness inside symbolic and bridge-verification envelopes, not exhaustive large-dimensional benchmarking. |

The comparison set should therefore be read as layered evidence rather than as a single universal leaderboard. Direct ANF, ESOP, ABC/BDD, and related representation baselines test matched bit-flip oracle resource efficiency; SSHR tests whether the proposed search remains competitive against a CNOT-oriented geometric method on small functions; mockturtle, CirKit, ROS-style LUT, and RevKit probes test whether the advantage persists outside the project’s own implementations; and internal ablations test whether the search controls contribute beyond deterministic construction. The results below use each family only for the claim it can support.

The experiments were executed as a reproducible artifact package rather than as isolated hand runs. Table 9 records the local compute envelope, manifest-level parallelism, artifact counts, and external toolchain provenance. Runtime numbers should be read within this workstation context; the logical resource counts are the portable experimental outcome.

Table 9: Compute and reproducibility audit. The table documents the workstation, parallel-run manifests, artifact coverage, and external-tool commits used by the current submission package.

| Aspect                   | Evidence                                                                                                                                                                                                                | Boundary                                                                                          |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Host compute envelope    | Apple M4 Pro; 14 physical/14 logical CPU cores; 24.0 GiB RAM; Apple M4 Pro, 20 GPU cores, Metal 4; Python 3.11.15                                                                                                       | Workstation context for reproducibility; not a hardware-mapping result.                           |
| Learning accelerator     | PyTorch 2.12.0; MPS=True; CUDA=False                                                                                                                                                                                    | GPU/MPS is available for neural training; synthesis resources remain logical-layer estimates.     |
| Parallel-run provenance  | 54 manifests record worker counts; max workers=10; traditional resource 10 workers/1770 rows; RevKit CLI 8 workers/708 rows; ROS-style LUT 8 workers/927 rows; CirKit 8 workers/177 rows; mockturtle 4 workers/177 rows | Wall times are machine-dependent and are not claimed as portable speedups.                        |
| Artifact coverage        | 87 top-level run/train/analyze scripts; 144 raw CSVs; 199 summaries; 98 manifests; 159 paper tables; 7 figure panels (21 PDF/PNG/SVG assets)                                                                            | Counts describe the current artifact package, not independent datasets.                           |
| External-tool provenance | mockturtle 25beb0e; CirKit 4531533; RevKit CLI 104eb35                                                                                                                                                                  | Toolchain rows are logic-level probes or exact-oracle reversible probes, not full hardware flows. |

Table 10 records the claim-to-artifact chain for the submission package. It is not a new result table; its role is to make each headline claim auditable by pointing to the scripts, CSVs, tables, figures, and manuscript anchors that carry the evidence.

Table 10: Submission traceability audit. Each row links a paper-facing claim family to the manuscript location and artifact chain used to support it.

| Claim family                     | Paper-facing use                                                                                                 | Manuscript anchor                                             | Boundary                                                                                                                 |
|----------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Method formulation               | Defines Resource-NMCTS as a logical ANF/FPRM search workflow.                                                    | Method, Table method-workflow                                 | Establishes the executable workflow, not a hardware compiler.                                                            |
| Contribution mapping             | Links each stated contribution to implementation, evidence, and claim boundary.                                  | Introduction, Table contribution-map                          | Prevents the introduction from making unsupported novelty claims.                                                        |
| Primary small-function resources | Supports T-count and weighted-score claims on matched $n \leq 6$ functions.                                      | Results, traditional functions                                | Same logical cost model; not a CNOT-only or hardware-level claim.                                                        |
| Baseline scope and fairness      | Explains why each baseline family is comparable and which claim it can support.                                  | Experimental Design, baseline matrices and counterpoint audit | Treats comparisons as layered evidence and reports unfavorable metric counterpoints rather than a universal leaderboard. |
| External toolchain probes        | Traces ROS-style LUT, mockturtle, CirKit, and RevKit CLI comparison evidence.                                    | Results, external probes                                      | Logic-level probes or exact-oracle reversible probe; not full hardware mapping.                                          |
| Multi-resource tradeoff          | Audits whether weighted-score wins imply raw-resource dominance.                                                 | Results, raw multi-resource dominance                         | Dominance uses T-count, CNOT, depth, and peak ancilla, not weighted score.                                               |
| Learned-control contribution     | Separates search-control baselines and promoted learned controllers from limited diagnostics.                    | Results, search contribution and ablation                     | Supports guarded learned-control evidence, not a claim that RL alone explains all gains.                                 |
| High-dimensional verification    | Traces large symbolic checks, phase checks, and complete truth-table bridge slices.                              | Results, high-dimensional verification                        | Large rows are symbolic or bridge-slice verified, not exhaustive high-dimensional truth tables.                          |
| Archive package integrity        | Hashes stable submission payload groups and records archive boundaries.                                          | Experimental Design, archive manifest                         | Excludes terminal submission audits and compiled PDF from payload hashes to avoid self-reference.                        |
| Reproducibility package          | Documents compute envelope, worker manifests, artifact counts, raw rerun entry points, and availability section. | Experimental Design and Data and Code Availability            | Repository-relative package until an archival DOI or anonymous link is supplied.                                         |

The submission archive manifest in Table 11 adds a package-level check over the stable payload groups. It hashes source files, tables, figures, raw measurements, summaries, manifests, scripts, trained models, and local tool adapters while excluding terminal submission audits and the compiled PDF from the digest set. This exclusion avoids self-referential hashes because the final PDF contains the manifest table itself; PDF existence and page count are checked separately by the readiness audit.

Table 11: Submission archive manifest. Each row reports a stable payload group, the number of files found, missing expected files, a category digest, and the scope boundary for that group.

| Payload group      | Files | Missing | Digest       | Boundary                                                                                                                                                                                                                                                |
|--------------------|-------|---------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manuscript source  | 4     | 0       | 43f032f80cda | Compiled PDF is checked by readiness audit to avoid self-referential hashes.                                                                                                                                                                            |
| Paper tables       | 157   | 0       | 07859d169237 | Terminal submission audit tables are excluded from the stable payload digest.                                                                                                                                                                           |
| Submission figures | 28    | 0       | 824254679c95 | Includes generated figure assets and plotted source data, not raw benchmark reruns.                                                                                                                                                                     |
| Raw measurements   | 144   | 0       | 87ee477b5c7c | Raw CSVs are regenerated only by the heavier run scripts, not by the lightweight rebuild.                                                                                                                                                               |
| Derived summaries  | 347   | 0       | f8ac05d194d0 | Terminal submission/package outputs are excluded so this manifest remains stable.                                                                                                                                                                       |
| Run manifests      | 84    | 0       | 233f7cc09243 | Submission-level terminal manifests are excluded from this digest group.                                                                                                                                                                                |
| Scripts and docs   | 99    | 0       | b5df2d5e5eef | Documents reproducible entry points; raw sweeps still require their individual drivers.                                                                                                                                                                 |
| Models             | 20    | 0       | 4c5c2a1bfc19 | Includes trained local policy artifacts when present; model retraining is not part of the lightweight rebuild.                                                                                                                                          |
| External adapters  | 1     | 0       | de5aa3b78998 | Includes local adapter source files used for external toolchain probes, not vendored tool repositories.                                                                                                                                                 |
| Submission support | 11    | 0       | 0164dddbbcff | Includes public package README, author-input packet, artifact guide, cover-letter, declaration, checklist, reviewer-brief, editor-screening, venue-selection, and structured metadata templates; ignored generated private submission text is excluded. |

## 6 Results

### 6.1 Traditional functions

Figure 2 and Table 12 summarize the traditional  $n \leq 6$  slice. Pareto-Resource-NMCTS has the lowest mean T-count and weighted score among the shown internal baselines, reducing T-count by 72.25% and score by 67.80% relative to direct ANF synthesis. SSHR-H retains the lowest mean CNOT count, so the result is not a CNOT-only dominance claim.

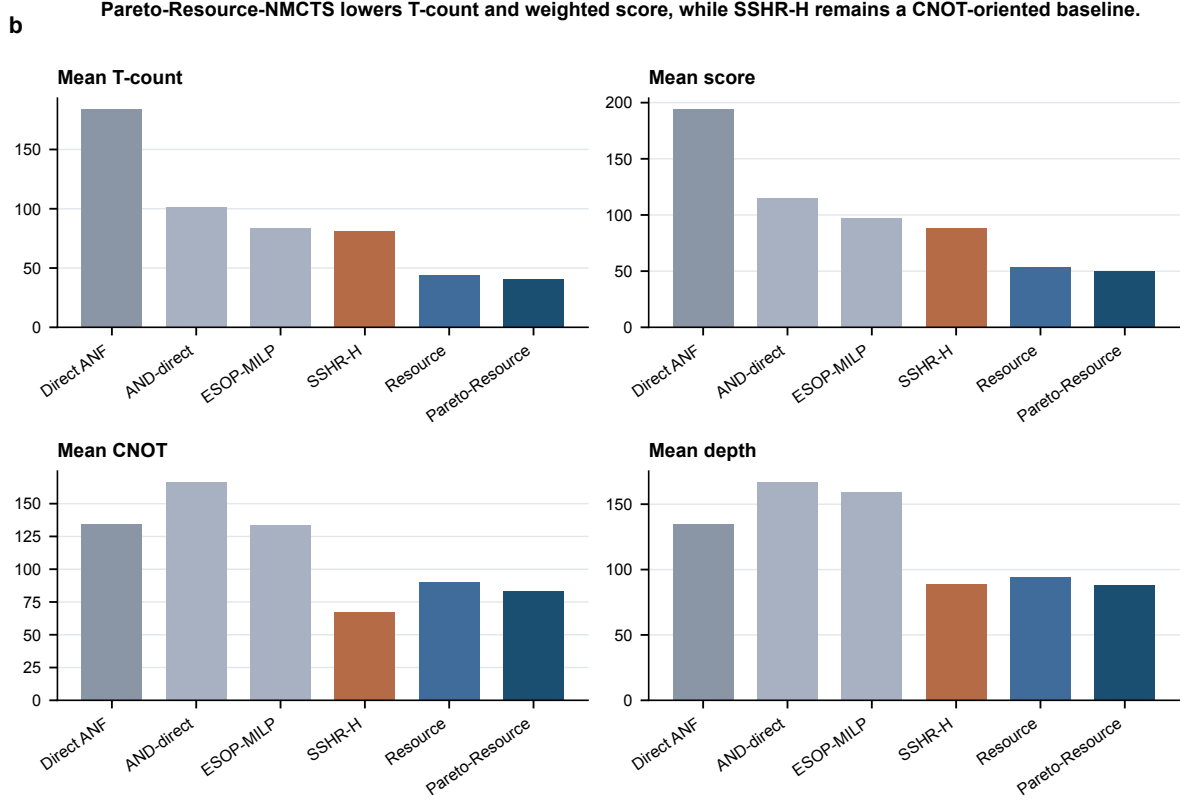


Figure 2: **Mean logical resources on the traditional slice.** Pareto-Resource-NMCTS has the lowest T-count and weighted score; SSHR-H remains a strong CNOT-oriented baseline.

Table 12: Mean logical resources on the  $n \leq 6$  traditional slice.

| Method                | Mean T      | Mean CNOT   | Mean depth  | Mean ancilla | Mean score  |
|-----------------------|-------------|-------------|-------------|--------------|-------------|
| Direct ANF            | 184.1       | 134.2       | 134.6       | 1.33         | 194.3       |
| AND-direct ANF        | 101.0       | 166.5       | 166.9       | 2.18         | 115.2       |
| ESOP-MILP             | 83.6        | 133.5       | 159.6       | 2.32         | 96.7        |
| SSHR-H                | 81.0        | <b>67.6</b> | 88.7        | 1.36         | 88.2        |
| Resource-NMCTS        | 43.9        | 89.9        | 94.5        | 1.92         | 53.2        |
| Pareto-Resource-NMCTS | <b>40.4</b> | 83.0        | <b>88.0</b> | 2.03         | <b>49.6</b> |

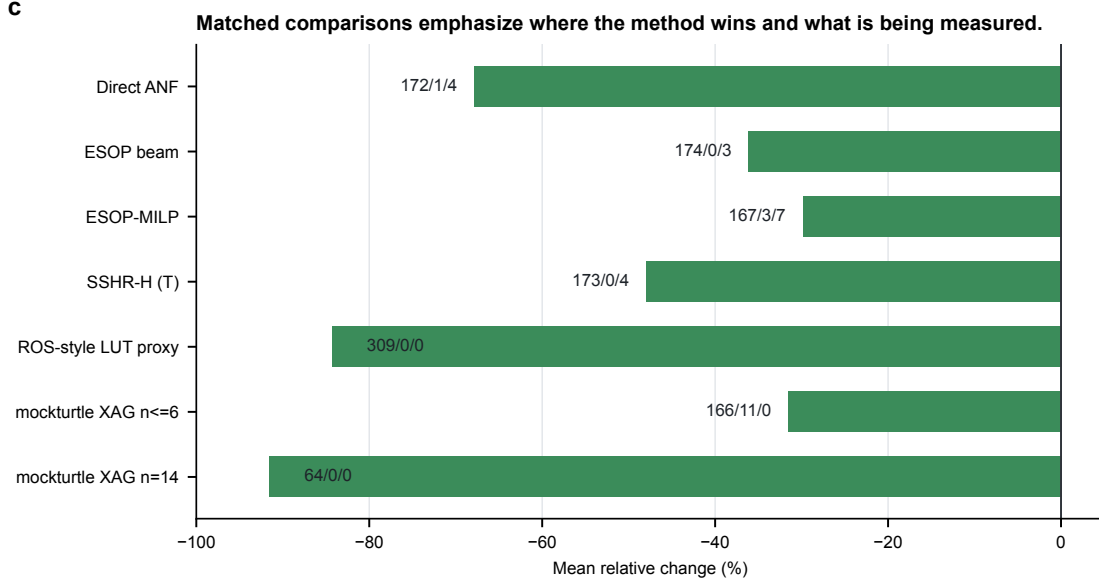
## 6.2 External logic-network and reversible-synthesis probes

Figure 3 summarizes paired score changes against several external or traditional baselines. The ROS-style LUT proxy covers 309 functions and 927 K-sweep rows, all truth-table checked; Pareto-Resource-NMCTS wins 309/309 score comparisons against the best-K proxy. The mockturtle probe uses official headers to resynthesize KLUT networks into XAGs; Pareto-Resource-NMCTS has 166/11/0 score wins on the traditional set and 64/0/0 on the  $n = 14$  set.

We further reselect the same verified LUT sweep under stricter line pressure. The minimum-ancilla selector reduces the LUT proxy’s peak ancilla by 32.47% relative to the best-score proxy, but increases its own weighted score by 40.67%. Pareto-Resource-NMCTS still obtains 309/0/0 score wins against this minimum-ancilla selector and against two line-weighted selectors. This result is a robustness check for LUT-parameter choice, not a claim that the official ROS SAT garbage-management flow has been reproduced.

Table 13: Line-aware reselection of the verified ROS-style LUT proxy sweep.

| Comparison                                                | Metric       | Items | W/L/T     | Mean $\Delta$ |
|-----------------------------------------------------------|--------------|-------|-----------|---------------|
| and_pareto_resource_nmcts vs external_ros_lut_min_ancilla | score        | 309   | 309/0/0   | -85.83%       |
| and_pareto_resource_nmcts vs external_ros_lut_min_ancilla | peak_ancilla | 309   | 301/0/8   | -68.21%       |
| and_pareto_resource_nmcts vs external_ros_lut_line_w10    | score        | 309   | 309/0/0   | -84.45%       |
| and_pareto_resource_nmcts vs external_ros_lut_k4_line_cap | score        | 309   | 309/0/0   | -85.02%       |
| external_ros_lut_min_ancilla vs external_ros_lut_proxy    | peak_ancilla | 309   | 197/0/112 | -32.47%       |
| external_ros_lut_min_ancilla vs external_ros_lut_proxy    | score        | 309   | 0/197/112 | +40.67%       |

Figure 3: **Paired baseline comparisons.** Negative values favor the target method. Labels show win/loss/tie counts.

The CirKit probe converts the same BLIF benchmarks to AIGER with ABC, applies CirKit AIG rewriting and multiplicative-complexity analysis, writes Verilog, and verifies the Verilog readback through ABC. It produces 177/177 correct traditional rows and 64/64 correct  $n = 14$  rows. Pareto-Resource-NMCTS wins all score comparisons but loses depth on most rows, making CirKit the strongest depth-oriented external probe in the current manuscript.

Table 14: CirKit 3 AIG/multiplicative-complexity probe on  $n \leq 6$  functions.

| Target vs CirKit AIG/MC   | Items | W/L/T    | Mean $\Delta$ |
|---------------------------|-------|----------|---------------|
| and_resource_nmcts        | 177   | 177/0/0  | -60.84%       |
| and_pareto_resource_nmcts | 177   | 177/0/0  | -62.34%       |
| and_fprm_polarity_archive | 177   | 177/0/0  | -58.96%       |
| and_direct_anf            | 177   | 124/53/0 | -16.74%       |
| direct_anf                | 177   | 46/131/0 | +35.23%       |
| and_esop_milp             | 177   | 150/27/0 | -39.41%       |
| sshr_h                    | 177   | 164/13/0 | -32.83%       |

Table 15: CirKit 3 AIG/multiplicative-complexity probe on the  $n = 14$  set.

| Target vs CirKit AIG/MC    | Items | W/L/T  | Mean $\Delta$ |
|----------------------------|-------|--------|---------------|
| and_resource_nmcts         | 64    | 64/0/0 | -94.42%       |
| and_profile_resource_nmcts | 64    | 64/0/0 | -94.42%       |
| and_pareto_resource_nmcts  | 64    | 64/0/0 | -94.46%       |
| and_fprm_linear_pair       | 64    | 64/0/0 | -94.27%       |
| and_fprm_root_beam         | 64    | 64/0/0 | -94.11%       |
| and_direct_anf             | 64    | 64/0/0 | -91.76%       |
| direct_anf                 | 64    | 64/0/0 | -86.22%       |
| and_fprm_greedy            | 64    | 64/0/0 | -94.08%       |
| and_affine_greedy          | 64    | 64/0/0 | -94.08%       |

The RevKit CLI probe supplies a closer reversible-synthesis comparison. Each function is embedded as the exact oracle permutation  $(x, y) \mapsto (x, y \oplus f(x))$ , and legacy RevKit reads the SPEC permutation before running TBS, DBS, and RMS synthesis. All 531 direct flow rows return. The per-function best-score RevKit portfolio is strong on line count, but Pareto-Resource-NMCTS retains 173/0/4 score wins and a 67.28% mean score reduction.

Table 16: Legacy RevKit CLI reversible-synthesis portfolio on  $n \leq 6$  functions. Each function uses the best-score result among TBS, DBS, and RMS.

| Target vs RevKit CLI best-score | Items | W/L/T   | Mean $\Delta$ |
|---------------------------------|-------|---------|---------------|
| and_resource_nmcts              | 177   | 173/0/4 | -66.32%       |
| and_pareto_resource_nmcts       | 177   | 173/0/4 | -67.28%       |
| and_fprm_polarity_archive       | 177   | 173/0/4 | -64.24%       |
| and_direct_anf                  | 177   | 167/6/4 | -32.79%       |
| direct_anf                      | 177   | 87/86/4 | +5.78%        |
| and_esop_milp                   | 177   | 172/1/4 | -51.50%       |
| sshr_h                          | 177   | 170/7/0 | +83.84%       |

### 6.3 Paired statistical evidence

The headline reductions above are paired by function or item, but mean relative improvements alone can hide skewed instances. We therefore recompute the central score comparisons directly from usable raw rows and apply an exact two-sided sign test that ignores ties. Table 17 shows that the main traditional, external-toolchain, and high-dimensional score claims have negative median relative changes as well as strongly one-sided win counts. The high-dimensional root-beam and fast-pair margins are smaller in magnitude than the external-probe margins, but they remain consistent across paired rows.

Table 17: Paired statistical evidence for score comparisons. Rows are recomputed from correct, non-skipped raw CSV entries matched by item name. Negative mean and median changes favor the target method;  $\log_{10} p$  is from a two-sided exact sign test over wins and losses, ignoring ties.

| Comparison                       | Pairs | W/L/T    | Mean $\Delta$ | Median $\Delta$ | $\log_{10} p$ |
|----------------------------------|-------|----------|---------------|-----------------|---------------|
| $n \leq 6$ Pareto vs direct ANF  | 177   | 172/1/4  | -67.80%       | -71.37%         | -49.54        |
| $n \leq 6$ Pareto vs ESOP beam   | 177   | 174/0/3  | -36.09%       | -32.56%         | -52.08        |
| $n \leq 6$ Pareto vs ESOP-MILP   | 177   | 167/3/7  | -29.84%       | -23.69%         | -44.96        |
| $n \leq 6$ Pareto vs SSHR-H      | 177   | 173/4/0  | -41.06%       | -43.05%         | -45.37        |
| $n \leq 6$ Pareto vs SSHR-I CNOT | 177   | 173/4/0  | -31.89%       | -33.91%         | -45.37        |
| $n \leq 6$ Pareto vs ABC-XAG     | 177   | 177/0/0  | -65.58%       | -65.25%         | -52.98        |
| ROS-style LUT best-K             | 309   | 309/0/0  | -84.27%       | -81.38%         | -92.72        |
| mockturtle XAG $n \leq 6$        | 177   | 166/11/0 | -31.50%       | -32.43%         | -35.96        |
| mockturtle XAG $n = 14$          | 64    | 64/0/0   | -91.49%       | -95.50%         | -18.96        |
| CirKit AIG/MC $n \leq 6$         | 177   | 177/0/0  | -62.34%       | -62.35%         | -52.98        |
| CirKit AIG/MC $n = 14$           | 64    | 64/0/0   | -94.46%       | -96.10%         | -18.96        |
| RevKit CLI exact oracle          | 177   | 173/0/4  | -67.28%       | -70.18%         | -51.78        |
| $n = 14$ Pareto vs root beam     | 64    | 60/0/4   | -6.31%        | -4.68%          | -17.76        |
| $n = 16$ Resource vs root beam   | 24    | 23/0/1   | -4.36%        | -3.36%          | -6.62         |
| $n = 18$ Resource vs fast pair   | 12    | 12/0/0   | -3.55%        | -1.79%          | -3.31         |

To avoid treating each mean effect as a single point estimate, we also compute nonparametric bootstrap intervals on the same matched rows. This uncertainty check does not introduce new measurements; it asks whether the observed effect sizes remain separated from zero when functions or items are resampled with replacement. Table 18 shows that the main small-function and external-toolchain comparisons remain negative with narrow intervals, while the high-dimensional root-beam and fast-pair comparisons keep smaller but still negative intervals. The table therefore strengthens the paired evidence without converting a weighted-score result into a universal resource-dominance claim.

Table 18: Bootstrap uncertainty intervals for paired score-effect estimates. Rows reuse the matched pairs from Table 17. Intervals are percentile bootstrap intervals over relative score changes; negative values favor the target method.

| Comparison                       | Pairs | Mean $\Delta$ [95% CI]   | Median $\Delta$ [95% CI] | Item p10/p90  |
|----------------------------------|-------|--------------------------|--------------------------|---------------|
| $n \leq 6$ Pareto vs direct ANF  | 177   | -67.80% [-69.91, -65.42] | -71.37% [-73.09, -69.62] | -78.88/-54.46 |
| $n \leq 6$ Pareto vs ESOP beam   | 177   | -36.09% [-38.67, -33.69] | -32.56% [-35.11, -29.81] | -55.01/-19.65 |
| $n \leq 6$ Pareto vs ESOP-MILP   | 177   | -29.84% [-32.70, -27.04] | -23.69% [-26.68, -21.47] | -62.94/-10.26 |
| $n \leq 6$ Pareto vs SSHR-H      | 177   | -41.06% [-43.38, -38.77] | -43.05% [-44.08, -39.82] | -55.50/-22.11 |
| $n \leq 6$ Pareto vs SSHR-I CNOT | 177   | -31.89% [-33.80, -29.90] | -33.91% [-36.68, -33.23] | -46.91/-14.30 |
| $n \leq 6$ Pareto vs ABC-XAG     | 177   | -65.58% [-67.10, -64.11] | -65.25% [-66.60, -63.34] | -76.73/-54.77 |
| ROS-style LUT best-K             | 309   | -84.27% [-85.73, -82.79] | -81.38% [-83.16, -79.32] | -99.62/-68.49 |
| mockturtle XAG $n \leq 6$        | 177   | -31.50% [-34.31, -28.62] | -32.43% [-33.58, -29.70] | -49.55/-9.96  |
| mockturtle XAG $n = 14$          | 64    | -91.49% [-94.09, -88.33] | -95.50% [-97.12, -93.60] | -98.93/-83.64 |
| CirKit AIG/MC $n \leq 6$         | 177   | -62.34% [-64.10, -60.59] | -62.35% [-64.66, -61.06] | -76.91/-47.69 |
| CirKit AIG/MC $n = 14$           | 64    | -94.46% [-95.72, -93.16] | -96.10% [-97.33, -94.71] | -98.96/-87.43 |
| RevKit CLI exact oracle          | 177   | -67.28% [-69.11, -65.23] | -70.18% [-71.53, -68.21] | -78.57/-57.54 |
| $n = 14$ Pareto vs root beam     | 64    | -6.31% [-7.89, -4.95]    | -4.68% [-5.21, -3.76]    | -13.44/-2.05  |
| $n = 16$ Resource vs root beam   | 24    | -4.36% [-6.61, -2.89]    | -3.36% [-4.22, -2.46]    | -6.53/-1.55   |
| $n = 18$ Resource vs fast pair   | 12    | -3.55% [-5.66, -1.81]    | -1.79% [-4.50, -1.29]    | -8.59/-1.24   |

## 6.4 Raw multi-resource dominance

Because Eq. (3) is only a ranking objective, we also test dominance without using the weighted score. A method is said to dominate another row only if it is no worse in T-count, CNOT count, logical depth, and peak ancilla, and is strictly better in at least one of them. Table 19 shows the resulting four-resource dominance counts for Pareto-Resource-NMCTS on the traditional slice. The method strongly dominates ESOP beam and many ESOP-MILP rows, but most comparisons against SSHR, CirKit, mockturtle, and RevKit are incomparable. This confirms

that those baselines occupy real CNOT, depth, or line-count tradeoff points even when their weighted scores are worse.

Table 19: Raw four-resource dominance of Pareto-Resource-NMCTS on  $n \leq 6$  functions. Dominance uses only T-count, CNOT count, logical depth, and peak ancilla; weighted score is not part of the predicate.

| Baseline                | Pairs | Pareto dom. | Base dom. | Incomp. | Equal |
|-------------------------|-------|-------------|-----------|---------|-------|
| Direct ANF              | 177   | 69          | 1         | 103     | 4     |
| ESOP beam               | 177   | 165         | 0         | 9       | 3     |
| ESOP-MILP               | 177   | 123         | 2         | 45      | 7     |
| SSHR-H                  | 177   | 33          | 4         | 140     | 0     |
| SSHR-I CNOT             | 177   | 9           | 3         | 165     | 0     |
| ABC-XAG                 | 177   | 33          | 0         | 144     | 0     |
| mockturtle XAG          | 177   | 11          | 0         | 166     | 0     |
| CirKit AIG/MC           | 177   | 21          | 0         | 156     | 0     |
| RevKit CLI exact oracle | 177   | 3           | 0         | 170     | 4     |

The same predicate can be applied to the whole method pool. Table 20 counts how often each method remains non-dominated by any other listed method on the same function. Pareto-Resource-NMCTS is non-dominated on 170/177 functions, while SSHR-I, mockturtle, and RevKit remain frequently non-dominated because they keep distinct resource tradeoffs.

Table 20: Non-dominated counts in the 12-method  $n \leq 6$  pool.

| Method                  | Rows | Non-dominated | Dominated |
|-------------------------|------|---------------|-----------|
| Pareto-Resource-NMCTS   | 177  | 170           | 7         |
| SSHR-I CNOT             | 177  | 162           | 15        |
| mockturtle XAG          | 177  | 159           | 18        |
| RevKit CLI exact oracle | 177  | 141           | 36        |
| Resource-NMCTS          | 177  | 124           | 53        |
| CirKit AIG/MC           | 177  | 56            | 121       |
| SSHR-H                  | 177  | 46            | 131       |
| ESOP-MILP               | 177  | 44            | 133       |
| ABC-XAG                 | 177  | 15            | 162       |
| Direct ANF              | 177  | 7             | 170       |
| ESOP beam               | 177  | 6             | 171       |
| AND-direct ANF          | 177  | 4             | 173       |

## 6.5 Search contribution and ablation

Table 21 consolidates the contribution analysis. The largest pre-portfolio gain comes from affine and Boolean-ring structural search. The learned prior gives smaller but non-degrading quality gains on the traditional slice, while the frontier policies mainly control the quality/time frontier in high-dimensional term-set experiments. Table 23 isolates the high-dimensional Boolean-ring and Boolean-screen branch. This table is deliberately mixed: it shows quality-seeking structural guards, a screen gate that preserves the selected resource vector while saving time, and a negative control where a fast screen is not resource-competitive. This prevents the structural-search claim from being reduced to a neural-prior claim or overstated as a universal screen improvement. Table 24 then compresses the expensive depth-frontier itself. On the listed scale and truth-bridge slices, evaluating only depth 2 and depth 4 exactly matches the full depth-2/3/4 frontier while removing 24.94–28.88% of frontier evaluation time. Table 25 adds a learned gate on top of this sparse frontier. Trained on generated  $n = 16, 20, 24$  term sets and applied unchanged to held-out  $n = 28, 40$  term sets, it runs depth 4 on 32/48 test pairs, records zero false skips, exactly matches the sparse-frontier score, and reduces sparse-frontier evaluation time by 17.39%. The deterministic sparse frontier remains the quality reference; the learned component controls search budget after the depth-2 screen. Table 26 then audits the same trained gate without

retraining on three independent seeds over  $n = 24, 28, 32, 40$ . Across 144 additional pairs it records zero false skips, matches the deterministic sparse frontier on every row, and saves 13.43% of sparse-frontier evaluation time. Figure 4 then sweeps the deployment threshold on the same audit. The selected threshold lies inside a zero-false-skip plateau; the best zero-false-skip point in the sweep saves 14.92%, while allowing one false skip saves 15.49% with a 0.01% mean score gap. Table 27 adds the earlier staged teacher/guard. Its depth-4 trigger is selected only on the large-policy validation split; on the independent  $n = 24, 28, 32, 40$  scale rows it remains within 0.04% of all-depth score while reducing staged planning time by 25.43%.

Table 21: Matched contribution decomposition. Negative mean score changes favor the first method in each comparison.

| Component                                                               | Dataset              | Pairs | Score W/L/T | Mean $\Delta$ score |
|-------------------------------------------------------------------------|----------------------|-------|-------------|---------------------|
| Affine greedy vs fixed-coordinate MCTS                                  | ablation_affine      | 321   | 165/88/68   | -12.12%             |
| Neural refine over affine greedy                                        | ablation_affine      | 322   | 65/0/257    | -1.08%              |
| Guarded Affine-NMCTS over no-guard                                      | ablation_affine      | 322   | 88/0/234    | -1.74%              |
| Learned prior for Resource-NMCTS                                        | traditional_resource | 177   | 39/0/138    | -1.10%              |
| Pareto archive over Resource-NMCTS                                      | traditional_resource | 177   | 68/0/109    | -3.26%              |
| No-MCTS portfolio over heuristic-only                                   | trad. ablation       | 177   | 54/0/123    | -2.51%              |
| Resource-NMCTS over no-MCTS portfolio                                   | trad. ablation       | 177   | 54/0/123    | -1.44%              |
| Pareto Resource-NMCTS over no-MCTS portfolio                            | trad. ablation       | 177   | 106/0/71    | -4.69%              |
| Highdim no-MCTS portfolio over root beam                                | n14 ablation         | 16    | 14/0/2      | -6.25%              |
| Highdim no-MCTS portfolio over linear-pair                              | n14 ablation         | 16    | 14/0/2      | -3.08%              |
| Linear-pair guard vs root beam at $n=14$                                | highdim_resource     | 64    | 60/0/4      | -3.00%              |
| Recursive linear-pair guard vs root beam at $n=15$                      | n15 scale            | 32    | 30/0/2      | -5.28%              |
| Recursive linear-pair guard vs shallow linear-pair at $n=16$            | n16 scale            | 24    | 23/0/1      | -2.54%              |
| Recursive linear-pair guard vs root beam at $n=16$                      | n16 scale            | 24    | 23/0/1      | -4.31%              |
| Baseline-preserving AI guard vs deterministic recursive guard at $n=16$ | n16 scale            | 24    | 6/0/18      | -0.05%              |
| Fast linear-pair guard vs root beam at $n=18$                           | n18 stress           | 12    | 6/0/6       | -1.91%              |
| Resource-NMCTS vs fast linear-pair at $n=18$                            | n18 stress           | 12    | 12/0/0      | -3.55%              |

Table 22 restates the search-control comparisons in the form used for claim interpretation. The bit-flip branch is compared against heuristic-only, beam-only, strengthened no-MCTS portfolio, MCTS, Pareto archive, and learned-prior/no-prior controls. The only random control in the current package is the same-budget phase/ $R_z$  shortlist audit; it is therefore not used as a bit-flip random baseline. This keeps the learning claim narrow: search control adds measured quality or budget-control evidence, but the largest gains also depend on the algebraic action space and baseline-preserving guards.

Table 22: Search-control baseline audit. Rows state which search/control baseline is being tested and which stronger claim remains outside the evidence for that comparison.

| Layer                                  | Comparison                                                 | Pairs | Score W/L/T | Mean $\Delta$ score | Claim boundary                                                                          |
|----------------------------------------|------------------------------------------------------------|-------|-------------|---------------------|-----------------------------------------------------------------------------------------|
| search-space baseline                  | Affine greedy vs fixed-coordinate MCTS                     | 321   | 165/88/68   | -12.12%             | This is not a neural-only effect.                                                       |
| deterministic search controls          | No-MCTS portfolio over heuristic-only                      | 177   | 54/0/123    | -2.51%              | This row does not isolate MCTS.                                                         |
| deterministic search controls          | No-MCTS portfolio over beam-only                           | 177   | 106/1/70    | -8.36%              | Beam is a child/search baseline, not the full method.                                   |
| MCTS over deterministic portfolio      | Resource-NMCTS over no-MCTS portfolio                      | 177   | 54/0/123    | -1.44%              | The magnitude is incremental, not the whole resource drop.                              |
| Pareto search control                  | Pareto Resource-NMCTS over no-MCTS portfolio               | 177   | 106/0/71    | -4.69%              | Ancilla tradeoffs still need separate reporting.                                        |
| learned prior                          | Learned prior for Resource-NMCTS                           | 177   | 39/0/138    | -1.10%              | It is not a runtime claim and should not be promoted as the main source of improvement. |
| high-dimensional deterministic control | Highdim no-MCTS portfolio over root beam                   | 16    | 14/0/2      | -6.25%              | This is symbolic/high-dimensional evidence, not exhaustive truth-table optimality.      |
| phase random control                   | Diverse phase shortlist top-512 vs same-budget random mean | 38    | 17/0/21     | -0.012%             | This is a phase/Rz proxy control, not a bit-flip random baseline or rotation synthesis. |

Table 23: Boolean-ring structural evidence in high-dimensional slices. Negative mean changes favor the first method in each comparison. The  $n = 18$  row is kept as a speed-only negative control rather than promoted as a quality result.

| Slice    | Comparison                               | Pairs | Score W/L/T | Mean $\Delta$ score | Mean $\Delta$ time | Interpretation                                  |
|----------|------------------------------------------|-------|-------------|---------------------|--------------------|-------------------------------------------------|
| $n = 14$ | Boolean guard vs pairwise-wide           | 12    | 9/0/3       | -0.82%              | +32.80%            | quality gain; slower exhaustive guard           |
| $n = 16$ | Boolean guard vs pairwise-wide           | 24    | 14/0/10     | -0.34%              | +22.80%            | quality gain; slower guard                      |
| $n = 16$ | Boolean guard vs deterministic deep      | 24    | 18/0/6      | -0.52%              | +356.64%           | stronger quality branch; expensive              |
| $n = 16$ | Boolean-linear deep vs pairwise-wide     | 24    | 14/6/4      | -0.22%              | -72.31%            | fast structural variant with small quality gain |
| $n = 20$ | Resource update vs Boolean screen        | 6     | 5/0/1       | -4.52%              | +453.05%           | quality frontier; much slower                   |
| $n = 20$ | Screen gate vs full Resource             | 6     | 0/0/6       | +0.00%              | -75.58%            | safe skip: same resources, less planning time   |
| $n = 20$ | Recursive Boolean screen vs old Resource | 6     | 5/0/1       | -7.47%              | -7.32%             | large-n positive screen; ancilla tradeoff       |
| $n = 18$ | Recursive Boolean screen vs old Resource | 12    | 0/11/1      | +36.94%             | -97.90%            | speed-only negative control; not promoted       |

Table 24: Sparse depth-frontier audit. The sparse controller evaluates depth-2 and depth-4 Boolean-ring screens, skips depth 3, and is compared with the full measured depth-2/3/4 frontier in the same raw files.

| Source               | $n$ scope   | Pairs | Score W/L/T | Mean $\Delta$ score | Mean $\Delta$ time | Depth-4 vs depth-3 |
|----------------------|-------------|-------|-------------|---------------------|--------------------|--------------------|
| scale-frontier       | 20,28,40    | 72    | 0/0/72      | +0.00%              | -27.57%            | 44/0/28            |
| scale-generalization | 24,28,32,40 | 96    | 0/0/96      | +0.00%              | -28.17%            | 52/0/44            |
| truth-bridge-n23     | 23          | 6     | 0/0/6       | +0.00%              | -24.94%            | 5/0/1              |
| truth-bridge-n24     | 24          | 6     | 0/0/6       | +0.00%              | -28.88%            | 3/0/3              |
| truth-bridge-n25     | 25          | 6     | 0/0/6       | +0.00%              | -27.06%            | 4/0/2              |

Table 25: Learned sparse depth-4 gate. The gate predicts whether the depth-4 Boolean-ring screen should run after the depth-2 sparse-frontier state has been evaluated. The threshold is selected on validation data with a conservative false-skip guard and is applied unchanged to held-out  $n = 28, 40$  test rows.

| Split | Pairs | Run d4 | False skips | Score W/L/T vs sparse | Mean $\Delta$ score | Mean $\Delta$ time | Score W/L/T vs depth-2 |
|-------|-------|--------|-------------|-----------------------|---------------------|--------------------|------------------------|
| train | 96    | 78     | 0           | 0/0/96                | +0.00%              | -10.17%            | 75/0/21                |
| valid | 48    | 38     | 0           | 0/0/48                | +0.00%              | -12.01%            | 34/0/14                |
| test  | 48    | 32     | 0           | 0/0/48                | +0.00%              | -17.39%            | 23/0/25                |

Table 26: Independent-seed sparse depth-4 gate audit. The trained gate and its validation-selected threshold are reused without retraining on three new seeds. The table reports aggregate and per- $n$  comparisons against the deterministic sparse depth-2/4 frontier.

| Scope  | Pairs | Run d4 | False skips | Score W/L/T vs sparse | Mean $\Delta$ score | Mean $\Delta$ time | Score W/L/T vs depth-2 |
|--------|-------|--------|-------------|-----------------------|---------------------|--------------------|------------------------|
| all    | 144   | 106    | 0           | 0/0/144               | +0.00%              | -13.43%            | 94/0/50                |
| $n=24$ | 36    | 29     | 0           | 0/0/36                | +0.00%              | -9.88%             | 27/0/9                 |
| $n=28$ | 36    | 27     | 0           | 0/0/36                | +0.00%              | -12.98%            | 24/0/12                |
| $n=32$ | 36    | 23     | 0           | 0/0/36                | +0.00%              | -18.19%            | 21/0/15                |
| $n=40$ | 36    | 27     | 0           | 0/0/36                | +0.00%              | -12.66%            | 22/0/14                |

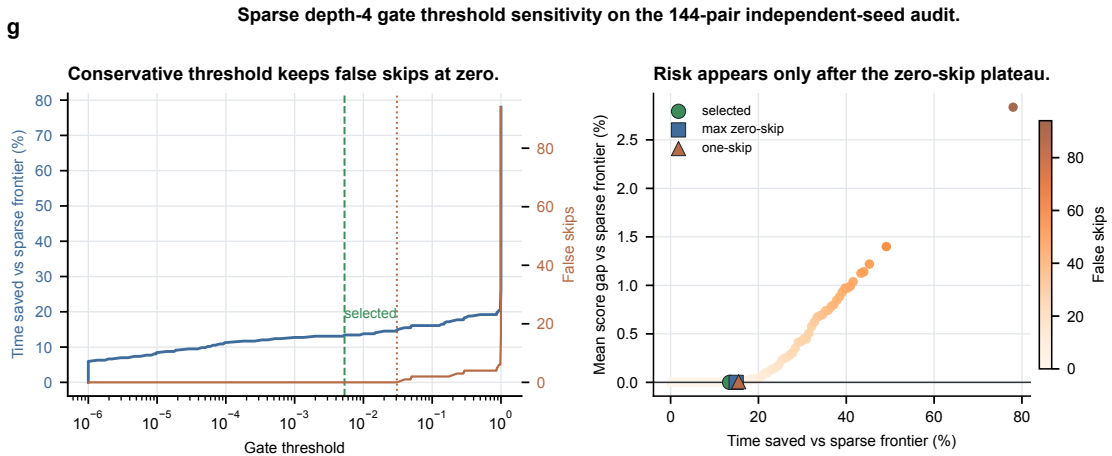


Figure 4: **Sparse depth-4 gate threshold sensitivity.** The sweep reuses the trained gate on the 144-pair independent-seed audit. The validation-selected threshold lies inside a zero-false-skip plateau, showing that the gate acts as a calibrated search-budget controller rather than as an ungarded depth-4 skip.

Table 27: Stage-gated depth-frontier controller. The depth-4 trigger is chosen on validation data and applied unchanged to scale and truth-bridge rows.

| Source               | Baseline              | Pairs | Score W/L/T | Mean $\Delta$ score | Mean $\Delta$ time | Run depth-4 |
|----------------------|-----------------------|-------|-------------|---------------------|--------------------|-------------|
| scale_generalization | adaptive_all_depth    | 96    | 0/4/92      | +0.04%              | -25.43%            | 52          |
| scale_generalization | screen_depth2         | 96    | 58/0/38     | -2.40%              | +893.25%           | 52          |
| scale_generalization | depth_frontier_policy | 96    | 5/3/88      | -0.06%              | +101.10%           | 52          |
| truth_bridge_n23     | adaptive_all_depth    | 6     | 0/0/6       | +0.00%              | -11.51%            | 5           |
| truth_bridge_n23     | screen_depth2         | 6     | 5/0/1       | -2.47%              | +1322.09%          | 5           |
| truth_bridge_n23     | depth_frontier_policy | 6     | 1/0/5       | -0.12%              | +94.20%            | 5           |

Table 28 separates promoted learned-control components from limited diagnostics. The frontier policies, sparse depth-4 gate, and phase shortlist provide the paper-facing learned-control evidence. In contrast, the current Boolean-neural guard and root-action neural ranker are kept as bounded diagnostics: they show small quality or evaluation-cost signals, but not enough quality/runtime advantage to carry the main claim.

Table 28: Learned-control evidence audit. Rows marked as limited are reported to bound the role of AI rather than promoted as headline improvements.

| Component                    | Scope                                      | Quality evidence                                         | Cost evidence                      | Paper role                               |
|------------------------------|--------------------------------------------|----------------------------------------------------------|------------------------------------|------------------------------------------|
| Depth-frontier policy        | held-out $n = 28, 40; 48$ rows             | vs oracle frontier 0/3/45, +0.04%                        | -51.30% time vs all-depth frontier | promoted quality/time selector           |
| Stage-gated frontier         | independent $n = 24, 28, 32, 40; 96$ rows  | vs all-depth 0/4/92, +0.04%                              | -25.43% staged planning time       | promoted validation-calibrated guard     |
| Sparse depth-4 gate          | multi-seed $n = 24, 28, 32, 40; 144$ pairs | vs sparse frontier 0/0/144, +0.00%; false skips 0        | -13.43% time vs sparse frontier    | promoted budget gate after depth-2       |
| Rank-diverse phase shortlist | held-out $n = 6$ phase search; 38 rows     | vs budget-32 17/0/21, -2.48%; vs wide-128 0/7/31, +0.00% | 512/8192 exact forms per function  | promoted phase-search pruning            |
| Boolean neural guard         | $n = 16$ high-dimensional guard; 24 rows   | vs deterministic 4/0/20, -0.12%                          | +94.49% runtime                    | limited quality guard, not runtime claim |
| Root-action neural ranker    | $n = 14$ root-action diagnostic; 10 rows   | vs beam4 +0.03%; oracle24 headroom -0.12%                | -98.06% ranking time vs beam4 eval | not promoted; future root-ranker target  |

Figure 5 visualizes the same conservative separation. Promoted controllers either preserve the reference score within a small tolerance or improve it, while reducing the measured planning time or exact-evaluation budget. The two limited diagnostics fall outside this quality-effort pattern and are therefore kept out of the headline claim.

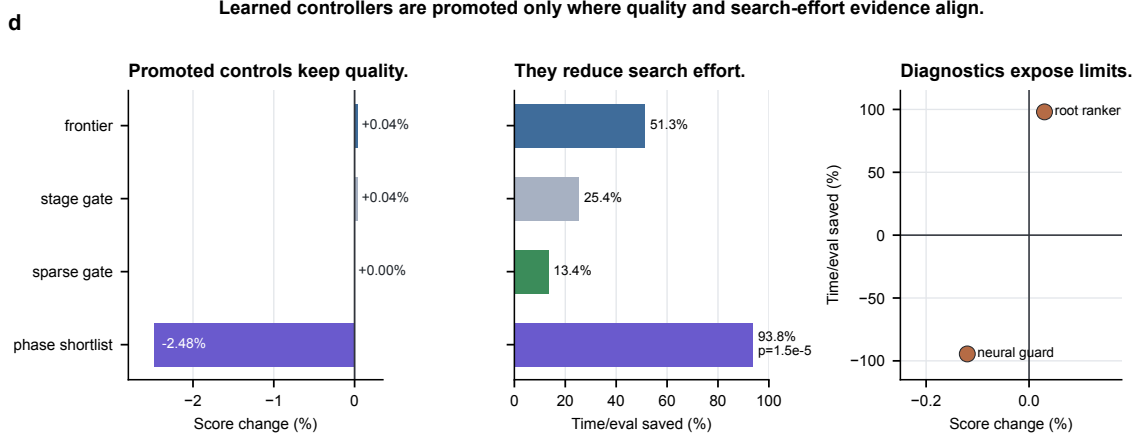


Figure 5: **Learned-control summary.** Promoted learned controllers are reported only where quality and search-effort evidence align. Positive time/evaluation saving means fewer frontier evaluations, less staged planning time, or fewer exact-scored phase candidates; negative score change favors the learned controller.

## 6.6 Score-weight robustness

The active score in Eq. (3) is intentionally a logical-resource ranking objective rather than a physical cost model. To check that the main comparisons are not artifacts of one coefficient choice, we recompute scores post hoc on verified rows under T-only, CNOT/depth-heavy, and ancilla-tight profiles. Table 29 shows that the small-function advantages over ESOP and SSHR-H and the high-dimensional advantages over root-beam or fast-pair guards survive these alternative profiles, while the CNOT/depth and ancilla-tight profiles expose the expected narrower margins.

Table 29: Post-hoc robustness to alternative logical-resource weights. Each cell reports score win/loss/tie counts and mean relative score change; negative values favor Resource-NMCTS or Pareto-Resource-NMCTS.

| Comparison                             | Paper score      | T-only            | CNOT-depth       | Ancilla-tight    |
|----------------------------------------|------------------|-------------------|------------------|------------------|
| $n \leq 6$ traditional: ESOP cube beam | 174/0/3, -36.09% | 174/0/3, -38.95%  | 174/0/3, -32.08% | 174/0/3, -32.75% |
| $n \leq 6$ traditional: ESOP-MILP      | 167/3/7, -29.84% | 166/0/11, -32.77% | 165/4/8, -25.44% | 167/3/7, -26.68% |
| $n \leq 6$ traditional: SSHR-H         | 173/4/0, -41.06% | 173/0/4, -47.93%  | 168/9/0, -27.87% | 172/5/0, -31.95% |
| $n = 14$ random ANF: FPRM root beam    | 60/0/4, -6.31%   | 60/0/4, -7.65%    | 60/0/4, -5.35%   | 56/4/4, -3.28%   |
| $n = 16$ random ANF: FPRM root beam    | 23/0/1, -4.36%   | 23/0/1, -4.70%    | 23/0/1, -4.28%   | 23/0/1, -3.43%   |
| $n = 18$ random ANF: FPRM root beam    | 12/0/0, -5.29%   | 12/0/0, -6.19%    | 12/0/0, -4.54%   | 11/1/0, -3.33%   |
| $n = 18$ random ANF: fast pair guard   | 12/0/0, -3.55%   | 12/0/0, -3.75%    | 12/0/0, -3.23%   | 12/0/0, -3.33%   |

## 6.7 Phase/Rz search

RevKit Python `oracle_synth` returns on all 177 traditional functions, but 171 rows contain non-Clifford  $R_z$  rotations, with 9242 such rotations in total. We therefore treat it as a phase/Rz lower-bound boundary rather than as an exact Clifford+T T-count comparison. The internal phase-parity emitter, fixed-polarity FPRM search, and affine FPRM phase search turn this boundary into a verifiable search branch.

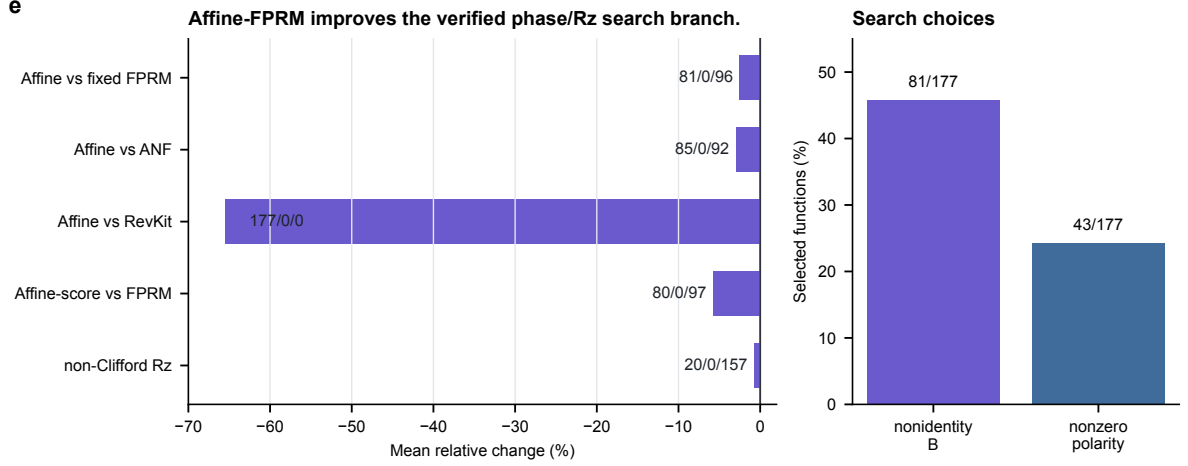


Figure 6: **Affine-FPRM phase search.** Affine preconditioning improves the phase-parity proxy over fixed-polarity FPRM and ANF baselines, while remaining a logical phase/Rz proxy rather than an approximate-rotation sequence.

Table 30: Affine-FPRM phase-parity search comparisons.

| Comparison                          | Items | W/L/T    | Mean $\Delta$ |
|-------------------------------------|-------|----------|---------------|
| Affine-FPRM- $T/R_z = 30$ vs FPRM   | 177   | 81/0/96  | -2.51%        |
| Affine-FPRM- $T/R_z = 30$ vs ANF    | 177   | 85/0/92  | -2.98%        |
| Affine-FPRM- $T/R_z = 30$ vs RevKit | 177   | 177/0/0  | -65.50%       |
| Affine-FPRM non-Clifford Rz vs FPRM | 177   | 20/0/157 | -0.73%        |
| Affine-FPRM-score vs FPRM           | 177   | 80/0/97  | -5.77%        |

Increasing the affine transform budget from 32 to 128 yields additional but bounded gains: under the  $T/R_z = 30$  proxy, the wide search gives 43/0/134 wins/losses/ties over budget 32 and reduces mean synthesis score by 0.60%. A rank-trained policy and a deterministic diversity reranker then compress this search. On the held-out  $n = 6$  split, the diverse top-512 policy exact-scores only 512 of 8192 affine forms per function, keeps the budget-32 comparison at 17/0/21 with a 2.48% mean reduction, and matches the wide-128 mean within 0.01%. Table 33 then uses the eight same-budget random repeats as a control. The independent unit is the held-out function: random repeats are averaged per function before the sign test. The strongest row is the diverse top-512 policy, which is 17/0/21 against the per-function random mean, has sign-test  $p = 1.53 \times 10^{-5}$ , and beats all eight random seed means. The absolute margins are small because the affine candidate space is dense, so this supports learned pruned phase-search feasibility rather than a claim that the current scorer solves phase/Rz synthesis globally.

Table 31: Affine-FPRM phase-search budget expansion.

| Metric              | Pairs | W/L/T    | Mean $\Delta$ |
|---------------------|-------|----------|---------------|
| score target        | 177   | 38/0/139 | -1.59%        |
| score+1/Rz target   | 177   | 40/0/137 | -0.93%        |
| $T/R_z = 30$ target | 177   | 43/0/134 | -0.60%        |
| non-Clifford Rz     | 177   | 3/0/174  | -0.29%        |
| total Rz            | 177   | 35/2/140 | -2.39%        |
| CNOT                | 177   | 37/2/138 | -1.74%        |
| depth               | 177   | 41/2/134 | -1.93%        |

Table 32: Rank-trained and diversity-reranked phase candidate pruning on held-out  $n = 6$  functions.

| Comparison                               | Items | W/L/T   | Mean $\Delta$ |
|------------------------------------------|-------|---------|---------------|
| Policy top-512 vs budget-32              | 38    | 17/0/21 | -2.40%        |
| Policy top-512 vs wide-128               | 38    | 0/9/29  | +0.13%        |
| Diverse policy top-512 vs budget-32      | 38    | 17/0/21 | -2.48%        |
| Diverse policy top-512 vs wide-128       | 38    | 0/7/31  | +0.00%        |
| Policy top-64 vs random top-64           | 38    | 10/7/21 | +1.37%        |
| Diverse policy top-64 vs random top-64   | 38    | 10/7/21 | +0.56%        |
| Policy top-128 vs random top-128         | 38    | 7/6/25  | +0.81%        |
| Diverse policy top-128 vs random top-128 | 38    | 8/4/26  | +0.03%        |
| Policy top-512 total Rz vs budget-32     | 38    | 15/0/23 | -4.57%        |
| Policy top-512 CNOT vs budget-32         | 38    | 16/1/21 | -2.80%        |
| Policy top-512 depth vs budget-32        | 38    | 16/1/21 | -3.29%        |

Table 33: Same-budget random-repeat control for the learned phase shortlist. Each row compares a learned top- $k$  shortlist with the mean of eight random top- $k$  shortlists on each held-out  $n = 6$  function before applying a paired sign test. Negative  $\Delta$  mean favors the learned shortlist.

| Policy  | Top- $k$ | W/L/T   | Sign $p$              | Mean target | Mean random | $\Delta$ mean |
|---------|----------|---------|-----------------------|-------------|-------------|---------------|
| Rank    | 64       | 16/3/19 | 0.0044                | 1482.48     | 1482.80     | -0.021%       |
| Diverse | 64       | 16/3/19 | 0.0044                | 1482.40     | 1482.80     | -0.027%       |
| Rank    | 128      | 14/5/19 | 0.0636                | 1482.29     | 1482.52     | -0.015%       |
| Diverse | 128      | 14/5/19 | 0.0636                | 1482.15     | 1482.52     | -0.025%       |
| Rank    | 256      | 12/6/20 | 0.2379                | 1482.14     | 1482.32     | -0.012%       |
| Diverse | 256      | 15/3/20 | 0.0075                | 1482.06     | 1482.32     | -0.018%       |
| Rank    | 512      | 14/3/21 | 0.0127                | 1482.03     | 1482.13     | -0.007%       |
| Diverse | 512      | 17/0/21 | $1.53 \times 10^{-5}$ | 1481.95     | 1482.13     | -0.012%       |

## 6.8 High-dimensional verification

Figure 7 summarizes the current verification envelope. The manuscript-level evidence includes 5640/5640 item-level symbolic runs for  $n = 20$ –40, 1512/1512 width-probe symbolic runs, 531/531 exact phase-search rows, 7611/7611 phase-policy pruning rows, and 400/400 complete truth table bridge method rows. The complete bridge spans  $n = 21$ –25 and checks the oracle relation, the expanded ANF plan, and the emitted-circuit symbolic semantics. Table 34 makes the large-scale evidence explicit by separating generated functions or settings from method rows and by reporting representative resource means only for the paper-facing method in each slice.

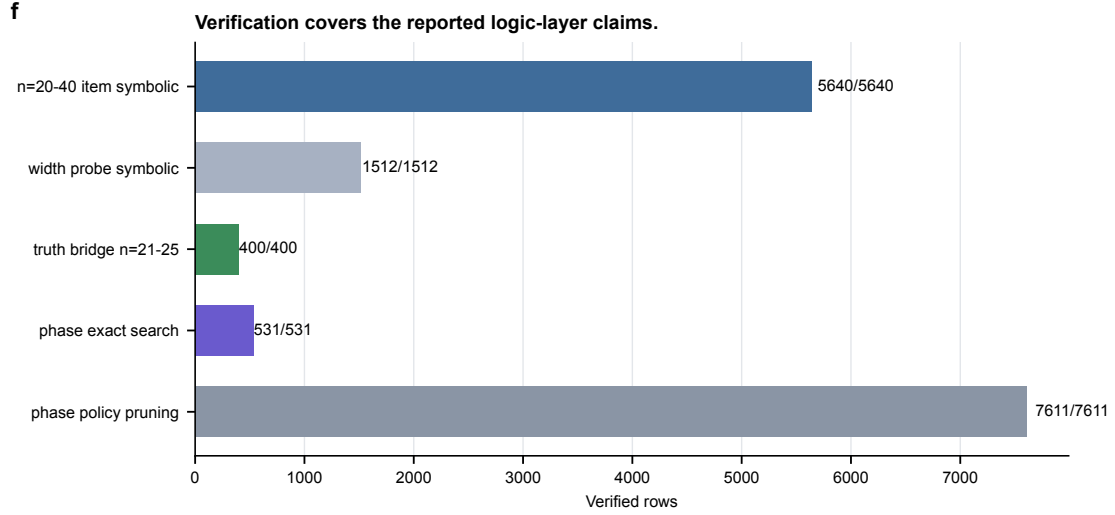


Figure 7: **Verification coverage.** Large term-set experiments are validated symbolically; phase-search and phase-policy rows are checked up to global phase;  $n = 21-25$  bridge slices add complete truth table checking on selected generated functions.

Table 34: Scaling and resource audit for high-dimensional experiments. The representative resource mean is method-specific; it is not an average over all baselines in the slice.

| Slice                       | $n$            | Fn./settings | Rows verified | Representative resource mean                                                          | Boundary                                                         |
|-----------------------------|----------------|--------------|---------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Large term-set frontier     | 24, 28, 32, 40 | 96           | 960/960       | score 1090.0; $T$ /CNOT/depth 986/1593/1593; anc. 5.3; time 7.40s                     | Term-set validation; no full truth table beyond bridge slices.   |
| Stage-gated frontier        | 24, 28, 32, 40 | 96           | 96/96         | score 1089.7; $T$ /CNOT/depth 985/1593/1593; anc. 5.3; time 10.57s                    | Controller audit; still term-set symbolic verification.          |
| Action-width probe          | 20, 28, 40     | 216          | 1512/1512     | score 1130.4; $T$ /CNOT/depth 1023/1648/1648; anc. 5.2; time 1.27s                    | Sensitivity check; wider root screens are not claimed as better. |
| Complete truth-table bridge | 21-25          | 40           | 400/400       | score 1126.2; $T$ /CNOT/depth 1019/1647/1647; anc. 5.3; time 4.16s; truth check 0.08s | Complete checking on generated bridge slices only.               |

Table 35: Verification scope and boundary.

| Check                                | Current coverage | Boundary                                                                               |
|--------------------------------------|------------------|----------------------------------------------------------------------------------------|
| $n = 20-40$ item-level symbolic runs | 5640/5640        | Not full truth-table enumeration.                                                      |
| Width-probe symbolic runs            | 1512/1512        | Supports the default action width, but remains term-set validation.                    |
| $n = 21-25$ complete bridge          | 400/400          | Small generated slices; not exhaustive coverage of all high-dimensional functions.     |
| Exact phase-search rows              | 531/531          | Verified up to global phase; no rotation-sequence output.                              |
| Phase-policy pruning rows            | 7611/7611        | Learned shortlist rows verified up to global phase; still no rotation-sequence output. |

## 7 Discussion

The evidence supports a specific, bounded claim: Resource-NMCTS is a logical-layer resource-constrained Boolean oracle synthesis method that improves T-count and weighted score over several fixed-representation and external-toolchain baselines. The claim is not that Resource-NMCTS dominates every metric. SSHR remains a strong CNOT-oriented small-function baseline; CirKit AIG/MC remains a strong depth-oriented external probe; and RevKit CLI uses fewer auxiliary lines on most traditional functions.

The ablations also calibrate the role of AI. The neural prior is useful, but the largest gain comes from the searchable algebraic action space and from guarded portfolio selection. This is a better fit for the evidence than claiming that deep reinforcement learning alone drives the result. The Boolean-ring structural audit further separates resource quality from planning cost: at  $n = 14$  and  $n = 16$  the guard improves score by 0.34–0.82% over the pairwise-wide baseline; at  $n = 20$  a deeper screen gives a 7.47% score reduction over the old Resource baseline with an ancilla tradeoff; and the  $n = 18$  screen-only row is much faster but worse in score. The sparse depth-frontier audit suggests a simpler planning-cost reduction inside the current frontier: depth 3 was never needed once depth 4 was measured in the audited slices, so a depth-2/4 frontier exactly matched all-depth resources while removing roughly one quarter of frontier evaluation time. This is an empirical controller for the present generated slices, not a proof that depth 3 is universally dominated. The learned sparse depth-4 gate adds a narrower AI-control result: after the sparse frontier has removed depth 3, the gate skips some depth-4 evaluations without changing score in either the held-out split or the independent multi-seed audit under its conservative threshold. The large, cost-aware, sparse-gated, and staged frontier controllers should be read as operating points on the same search frontier: quality-oriented single-shot selection, faster cost-aware selection, learned sparse-budget control, and a validation-calibrated staged teacher. In the phase/Rz branch, affine and polarity search give verified improvements, and the rank-trained diverse shortlist now tracks the wide-search frontier with 6.25% of the exact evaluations. However, random-repeat margins remain small, so this branch should be read as learned pruning evidence rather than as a solved rotation-synthesis policy. The next target is rotation-spectrum-aware optimization and sequence-level audit.

Several limitations are deliberate. All results are logical-layer estimates; the paper does not model routing, hardware connectivity, native gate sets, noise, or magic-state factories. The ROS-style LUT result is a proxy rather than a full ROS reproduction with SAT garbage management; the line-aware LUT reselection only tests parameter sensitivity within the same proxy. The RevKit CLI baseline is a real exact-oracle reversible-synthesis probe, but CNOT/depth are derived from RevKit’s Toffoli-control distribution and the experiment is still not a hardware mapping. High-dimensional symbolic checks validate emitted circuit semantics but do not replace full truth-table enumeration beyond the bridge slices.

## 8 Conclusion

We presented Resource-NMCTS, a neural Monte Carlo tree search framework for resource-constrained quantum Boolean oracle synthesis over verifiable ANF/FPRM actions. On matched small-function benchmarks, the method substantially lowers T-count and weighted score relative to direct ANF, ESOP, SSHR-H, and ESOP-MILP baselines. On external probes based on ROS-style LUT mapping, mockturtle, CirKit, and legacy RevKit CLI reversible synthesis, the method retains a strong weighted-score advantage while exposing clear depth and ancilla tradeoffs. Large symbolic experiments and  $n = 21$ – $25$  complete truth-table bridges support the semantic correctness of the logical-layer circuits. The sparse depth-frontier analysis further reduces high-dimensional planning cost by skipping depth 3 in the audited frontier, and the learned sparse depth-4 gate preserves sparse-frontier score across both held-out and independent-seed

audits while removing additional depth-4 evaluations. The staged frontier analysis remains a validation-calibrated controller when depth-4 evaluation itself should be conditional. The next step is not to claim hardware-level optimality, but to strengthen the phase/Rz policy and complete fuller toolchain reproductions while keeping the claim at the logical synthesis layer.

## Data and Code Availability

The artifact package used for this manuscript is maintained in the project repository under `resource_nmcts_experiment/`. The directory contains the synthesis code, neural and search-policy scripts, external-toolchain probe drivers, generated raw and summary CSV files, manifest JSON files, LaTeX table fragments, figure source data, and the compiled manuscript PDF. The main reproducibility entry points are the `run*.py`, `train*.py`, and `analyze*.py` scripts listed in the project README; generated measurements are stored in `results/raw*.csv`, `results/summary*.csv`, and `results/manifest*.json`. The paper tables are generated under `paper_latex/tables/`, and the submission figures and source data are stored under `paper_latex/figures/submission_v36/`. The lightweight derived submission package can be rebuilt with `./rebuild_submission_package.sh`; this regenerates paper-facing tables, figures, the submission archive manifest, the uploadable payload archive, audits, and the English PDF from existing experiment artifacts, but does not rerun raw benchmark sweeps, external-toolchain probes, or neural training jobs. The generated payload archive `submission_package/dist/resource_nmcts_submission_payload.tar.gz` is accompanied by a SHA256 sidecar and a JSON file manifest. The archive manifest records category-level hashes for stable payload groups; terminal submission audits and the compiled PDF are checked separately to avoid self-referential digests. The experiments use the `mcts-qoracle` Conda environment and the direct interpreter `/opt/anaconda3/envs/mcts-qoracle/bin/python`. External-tool rows are reported with manifest-level provenance; hardware mapping, routing, and magic-state-factory artifacts are intentionally absent because they are outside the logical-layer scope of the study.

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